

Pulse Plethysmogram Assignment

Medical instrumentation 1: Lab Project

Introduction

A pulse plethysmograph, also known as PPG, is a medical device used to measure the changes in blood volume by emitting light at a certain wavelength into the microvascular bed of tissue of a patient's finger. The light is absorbed by the blood vessels, and the change of light absorptions are directly related to the pulsatile flow of blood. Subsequently, the change in blood volume correlates with the patient's heart pulse.

Pulse plethysmographs are commonly used in medical contexts, such as hospitals and clinics, but can also be used at home. The device monitors blood pressure and cardiac output, as well as aid in the diagnosis of peripheral vascular diseases¹. PPGs help in detecting abnormalities in a patient's respiratory rate, blood pressure, ankle-brachial pressure, vascular aging, as well as in their neurological, cardiovascular, and nervous systems². For this reason, understanding the electronic functioning and mechanisms behind PPGs is important for the biomedical engineering community.

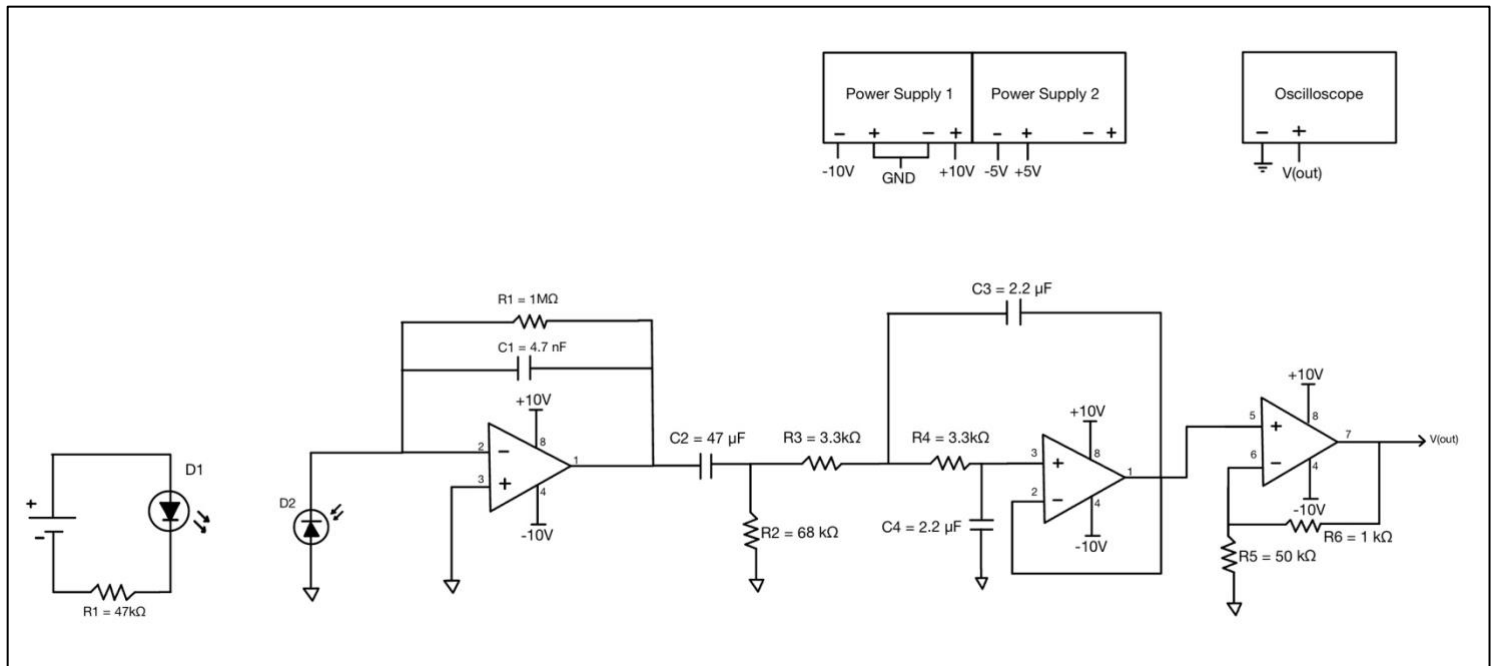
Method

Circuit diagram

To create the pulse plethysmogram, designing two main circuit blocks was necessary, the first being a LED driver, and the second being a signal conditioning block. The signal conditioning block was made of four components: a trans-impedance amplifier, a first order high pass filter, a Sallen-Key low pass filter, and a non-inverting amplifier. Below is a schematic of the full circuit diagram used (*breadboard circuit in appendix 1*).

¹ Allen, J. (2007). *Photoplethysmography and its application in clinical physiological measurement*. PubMed. National Library of Medicine. Available at: <https://pubmed.ncbi.nlm.nih.gov/17322588/#:~:text=The%20PPG%20technology%20has%20been,also%20detecting%20peripheral%20vascular%20disease>.

² Almarshad, A. Islam, S. Al-Ahmadi, S. and Bahammam, A. *Diagnostic features and potential applications of PPG signal in healthcare: a systematic review*. PubMed central. National Library of Medicine. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8950880/>



Design rationale

On the one hand, the LED driver circuit consisted of a resistor connected to an infra-red light emitting diode (TSAL6100) in series. The resistor's aim is to limit the current flowing through the LED, and therefore prevent it from burning. According to the absolute maximum ratings found on the diode's datasheet³, the maximum forward current that the diode can take is of 100 mA. Thus, for an input voltage of 5V, the resistor value had to be: $R = \frac{V}{I} = \frac{5}{100 \times 10^{-3}} = 50 \Omega$. The lab's closest resistor value was of 47Ω, and was therefore chosen instead.

On the other hand, the signal conditioning circuit's purpose was to convert the light into an output voltage, which was then filtered and amplified, before being converted into a digital signal through the Arduino device. The first component of the signal conditioning block is a trans-impedance amplifier. As mentioned above, its purpose is to convert the current induced by the infrared light emitted from the LED driver circuit into a voltage. The gain of the amplifier depends on the feedback resistor, which usually has a very large value. In this case a value of 1 MΩ was used. If the gain is too high, then the output voltage of the op-amp may have a DC offset. The resistor's goal is to prevent feeding too much current into the op-amp and burning it. Adding a feedback capacitor helps stabilize the signal, but also creates a simple first-order low pass filter. For a cut-off frequency of 30 Hz, and a resistor of 1 MΩ, a capacitor of 4.7 nF was chosen.

After obtaining a voltage signal, removing the offset was crucial. A first order high pass filter was used to do so. As a person's regular heart beats approximately 60 times per minute (once per second), filtering at around 1 Hz was appropriate. In fact, it can be found that a cut off-frequency

³ (2012). High Power Infrared Emitting Diode, 940 nm, GaAlAs, MQW. Vishay Semiconductors. Available at: https://moodle.ucl.ac.uk/pluginfile.php/6001787/mod_resource/content/0/2049675.pdf

of $f_c = 0.5 \text{ Hz}$ is adequate to separate DC component from the signal⁴. Using the formula of the cut-off frequency for a high-pass filter $f_c = \frac{1}{2\pi RC}$, a resistor value of $68 \text{ k}\Omega$ and a capacitor value of $47 \text{ }\mu\text{F}$ were chosen.

After removing the DC offset, a Sallen-Key low pass filter was added. Its purpose was to filter out noise from mains (muscle movement, motion artefacts). Similar to the high pass frequency, the cut-off frequency formula of $f_c = \frac{1}{2\pi RC}$ was used. For a cut-off frequency of $f_c = 20 \text{ Hz}$, values of $3300 \text{ }\Omega$ and $2.2 \text{ }\mu\text{F}$ were used for the resistors and capacitors respectively.

Subsequently, as the signal must have a voltage peak-per-peak value of at least 4 volts, a non-inverting amplifier was used to amplify the signal obtained. As the voltage recorded at the input of the non-inverting op-amp (i.e. pin 5) was around $v_{in} = 100 \text{ mV}$, using the transfer function of the non-inverting op-amp:

$$v_{out} = v_{in} \left(1 + \frac{R_2}{R_1}\right)$$

$$\frac{v_{out}}{v_{in}} - 1 = \frac{R_2}{R_1}$$

For $v_{out} = 5 \text{ V}$:

$$\frac{R_2}{R_1} = \frac{5}{100 \times 10^{-3}} - 1 = 49$$

Thus, a gain of around 49 was required. For this, from the resistors available in the lab, values of $R_2 = 47 \text{ k}\Omega$ and $R_1 = 1 \text{ k}\Omega$ were used.

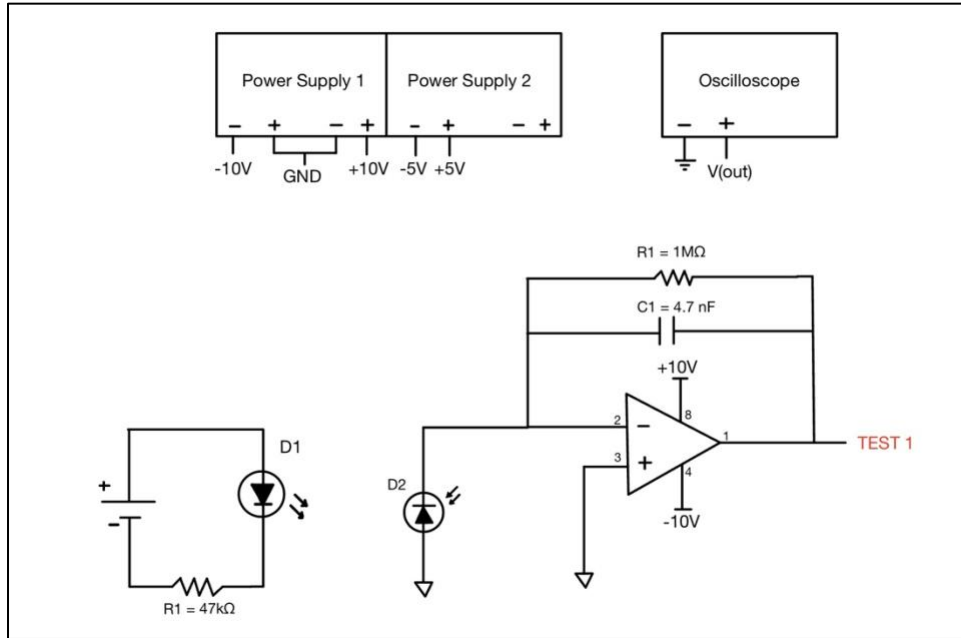
Finally, +10V and -10V were used to power all the op-amps.

Testing

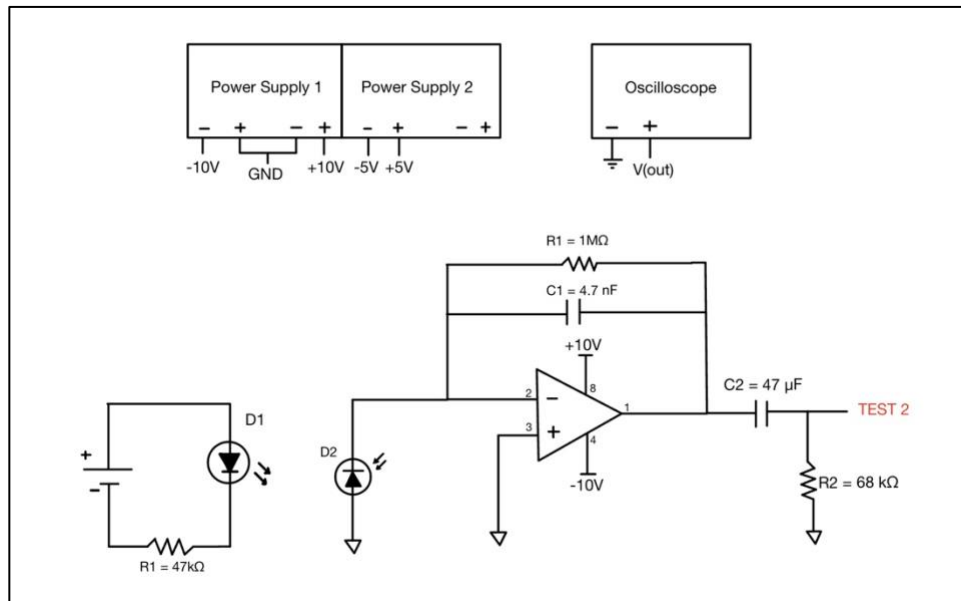
To test the functionality of the circuit, the measuring probe was connected to the output of every component and the ground (one component at a time).

The first step was ensuring that the photodiode was receiving the light and inducing an electrical current. This was checked by placing the probe at the pin 1 of the trans-impedance op-amp, and gently pressing the subject's finger on and off the photodiode. Below is the circuit diagram for the tested component.

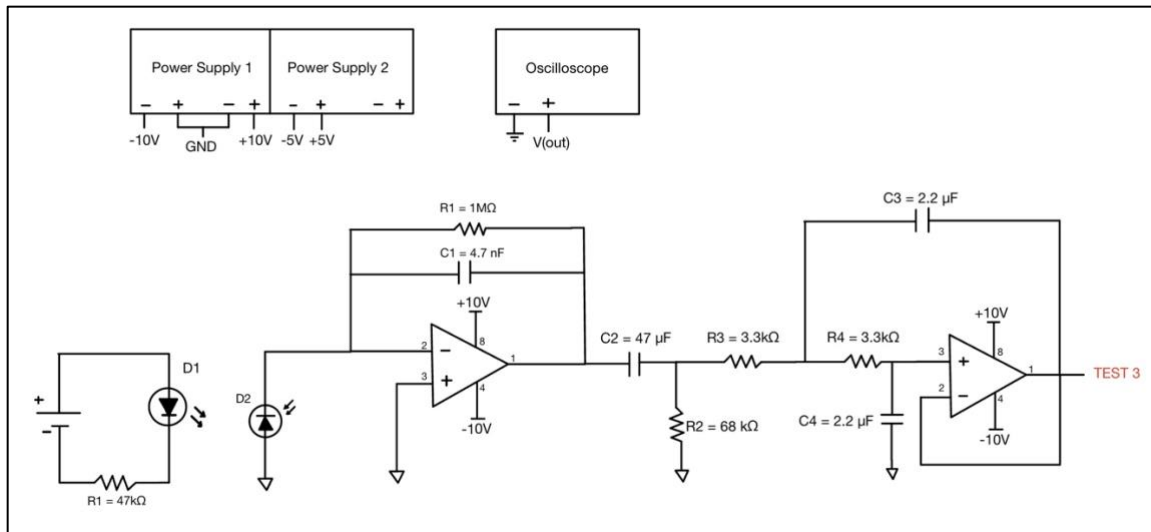
⁴ Kristan, P., Ferenets, R., Meigas, K. Lindberg, L.G., Temitski, K. Viigima, M. *New plethysmographic signal analysis algorithm for arterial stiffness estimation*. PubMed central. National Library of Medicine. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6426305/#:~:text=Because%20PPG%20is%20an%20indication,diseases%2C%20such%20as%20vascular%20aging>.



Once this step was completed, checking if the DC offset was eliminated was done by placing the measuring probe after the high pass filter (between the capacitor and resistor).



The third step was checking whether the Sallen-Key low pass filter was eliminating the noise of the signal. Thus, the probe was placed on pin 1 of the second op-amp.



The fourth and final step was to ensure the signal was amplified. The probe was therefore placed at the last output of the full circuit (i.e., pin 7 of the second op-amp) and measured its final output voltage.

Finally, after ensuring all components functioned properly, the code was tested by compiling it on Arduino and checking if there were any errors in the output window.

Results

By pressing and letting go of the photodiode with a subject's finger, the signal on the oscilloscope was increasing and decreasing, indicating that the photodiode was functional.

Then, after placing the probe on the output of the trans-impedance amplifier, the following signal was obtained:

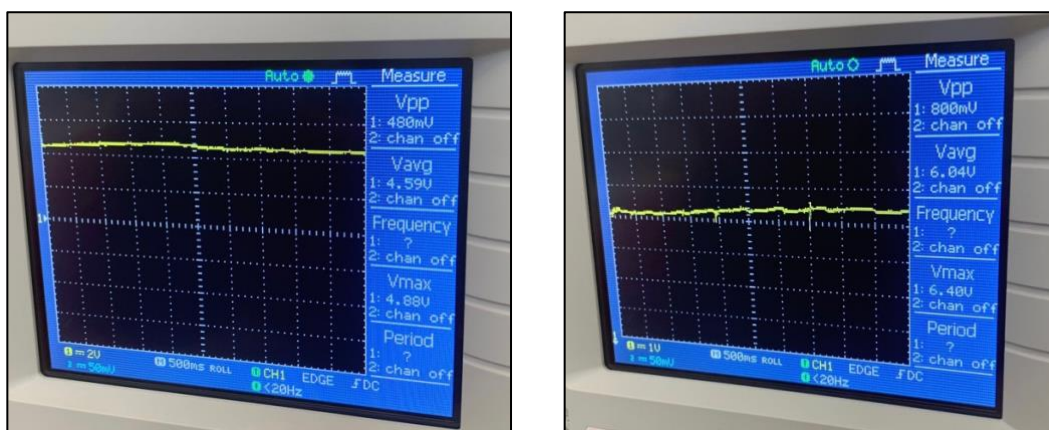


Figure 3.1. Pulse visible on the oscilloscope after the trans-impedance amplifier, on the left before adjustments and on the right after adjustments.

A periodic pulse signal was slightly visible, though lowering the volts/div in the range of millivolts was needed to see it better. As expected, a very high offset was noticeable: the image on the left

shows the signal without modifying the vertical position (reference is at 0), and the image on the right shows the signal after adjusting the vertical position and lowering the volts/div to 1V instead of 2V. Overall, however, the trans-impedance amplifier operated correctly. The signal obtained after the high-pass filter can be seen in figure 3.2 below.

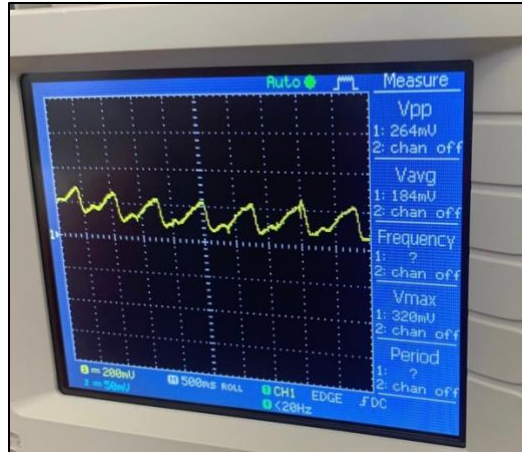


Figure 3.2. Pulse visible on the oscilloscope after the first-order high pass filter.

Here, the signal was much cleaner, and the DC offset was eliminated (no adjustments were needed). The pulse was in the 200 mV volts/div range and still had some noise. After the Sallen-key low pass filter:

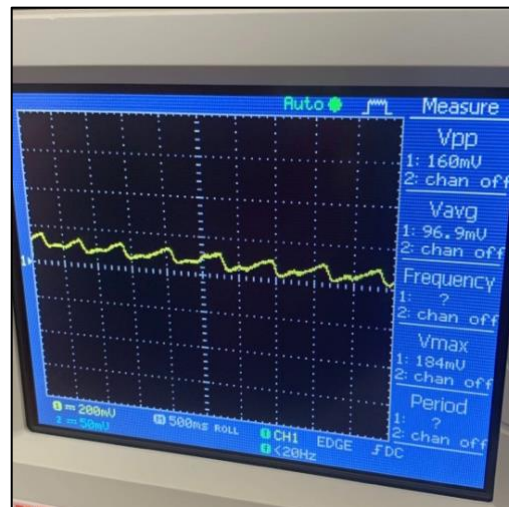


Figure 3.3. Pulse visible on the oscilloscope after the second-order Sallen-key low pass filter.

The pulse appeared clean and visible but had a very low amplitude. As measured on the oscilloscope, the signal at the output of the Sallen-key low pass filter was of around 160 mV. By amplifying it by a gain of around 47, it was expected to have a signal of around $\frac{160 \times 47}{1000} \approx 8 \text{ V}$. Thus, the final test was at the end of the signal conditioning block, after the non-inverting amplifier:

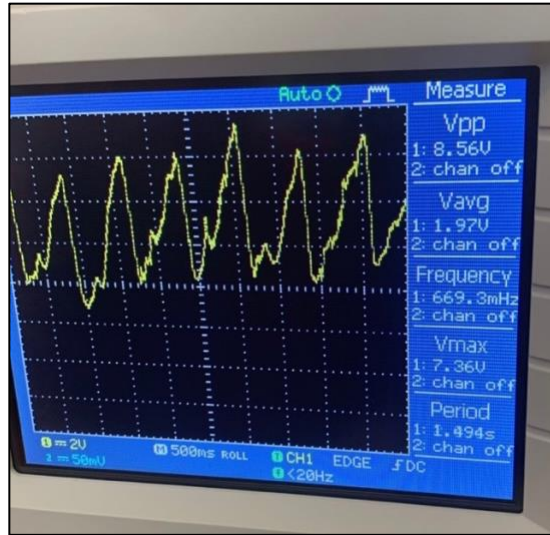


Figure 3.4. Pulse visible on the oscilloscope after the second-order Sallen-key low pass filter.

Figure 3.4. shows the final pulse obtained. The PPG signal (heart pulse) is in fact of around 8V ($8.56\text{V} > 4\text{ Vp-p}$), and has a low amount of noise, therefore meeting the requirement of the desired signal.

The last test was ensuring the analog signal was converted to a digital signal through Arduino. This was done by connecting the output of the signal conditioning block (pin 7 of non-inverting amplifier) to the analog pin A3 on Arduino uno, and connecting the ground of the circuit to GND. After uploading the code (*appendix 2*) to the Arduino device, the following signal was obtained on the Serial plotter:

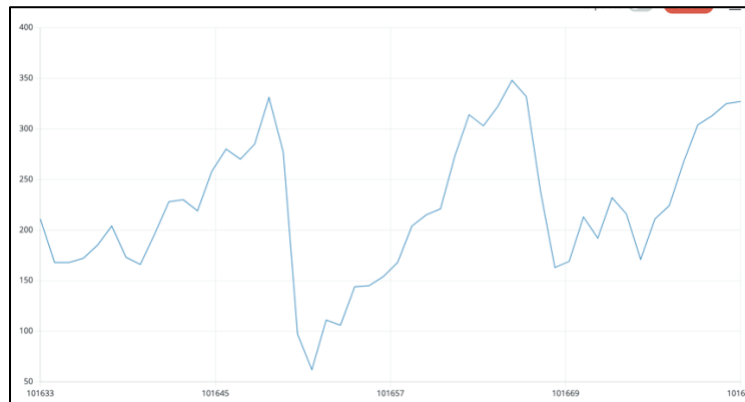


Figure 3.5. Plot showing the pulse visible on the computer after being processed by Arduino Uno.

Discussion and Conclusion

Although a pulse was visible on the oscilloscope, several improvements could have made the circuit more efficient.

To begin, the LED driver block could have been made with a BJT transistor instead of a simple LED and resistor in series. The transistor's aim is to take the voltage provided by the power supply and amplify it. This allows a much larger current to flow through the infrared light emitting diode than what the power supply could have provided itself. The higher the current through the LED, the more light can be emitted, and therefore the more accurate the pulse can be detected in the subject's finger.

Furthermore, it can be found that near-infrared diodes penetrate the skin the best⁵ (800 to 2,500 nm) and that the longer the wavelength the better the penetration through skin⁶. The diode used was the TSAL6100, with a wavelength of 940 nm. Finding a diode with a longer wavelength could therefore potentially improve the signal's accuracy.

Secondly, a first order high-pass filter was used. First order filters have a high sensitivity to non-idealities in the capacitors and component tolerances, they cannot provide a gain higher than 1, have suboptimal input and output impedances, and can cause loading effects when cascaded⁷. Thus, active filters - such as a second-order Sallen-Key high pass filter - should be used instead. This would allow an increase of the roll-off from 20 to 40 db./decade, as well as a better control of the sharpness of the filter. Overall, it would have filtered the noise better.

In fact, as a first order low pass filter was already built at the output of the trans-impedance amplifier, prioritizing a Sallen-Key high-pass rather than low pass would have been more efficient.

Thirdly, as mentioned above, both diodes were extremely sensitive to movement and noise. The subject needed to have warm hands – which was difficult due to the cold temperature in the room - and be very stable, as pulses were recorded only when the subject was not moving. Furthermore, the pulse appeared only in specific positions, angles, and worked better on some fingers than others, leading to a circuit which only functions in very particular cases.

In addition to the movement sensitivity, the photodiode was also sensitive to external light sources (lab room): when gently covering the diodes with a piece of cloth, pulses were much cleaner. Thus, in ideal settings, covering the circuit with a dark cloth would be suitable.

Finally, the Arduino signal that was displayed could have been read slower, so that more pulses were visible on the serial plotter. This means that the serial reading number could have been decreased to less than 1200. Moreover, a more sophisticated code which converts the pulse into bpm could have been implemented.

Overall, these arguments indicate that despite having a functioning PPG circuit and a relatively clean heart pulse signal, many improvements can still be made to make the circuit more effective before being able to use it for diagnosis in real medical settings.

⁵ (2021). *LED Light Therapy*. Cleveland Clinic. Available at: <https://my.clevelandclinic.org/health/treatments/22146-led-light-therapy#:~:text=Blue%20light%20affects%20the%20uppermost,Near%20infrared%20light%20penetrates%20deepest>.

⁶ Bezzant, J. L. *Penetration of human skin by ultraviolet light*. Available at: [https://library.med.utah.edu/kw/derm/pages/meet_2.htm#:~:text=The%20longer%20the%20wavelength%20of,\(320%20to%20400%20nanometers\)](https://library.med.utah.edu/kw/derm/pages/meet_2.htm#:~:text=The%20longer%20the%20wavelength%20of,(320%20to%20400%20nanometers)).

⁷ (2023). Lecture notes.

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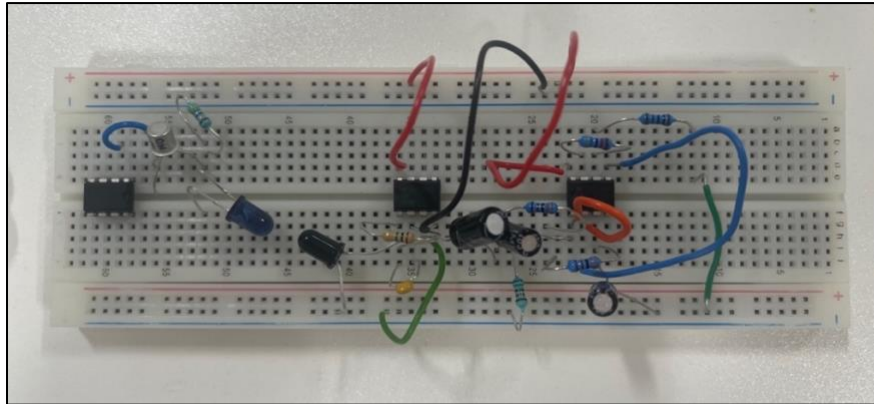
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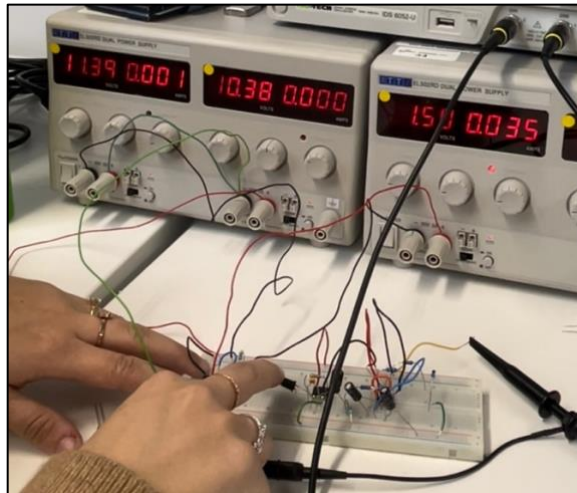
Appendix

Appendix 1

Circuit used during the project. *Note: a part of the BJT circuit diagram was built on the breadboard, however due to time limitations, it was not finished nor used.*



Power supply and oscilloscope connections, finger placement for measurement taking, as well as the probe connection to the output of a component and ground:



Appendix 2

Code used to convert the analog signal into a digital signal through Arduino:

```
1  int analogPin = A3; // output of the non-inverting amplifier connected to analog pin 3 on arduino board
2  int val = 0; // variable created to store the value read
3
4  void setup() {
5      Serial.begin(1200); // Setup serial, and set it to 1200 (to read the signal slower)
6  }
7
8  void loop() {
9      val = analogRead(analogPin); // Read the signal arriving in the input pin (A3)
10     Serial.println(val); // Debug and display the value on the serial plotter
11 }
```